

Abdullah Syed

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SUMMARY

Process Engineer with 2 years in flex/rigid PCB manufacturing (SMT and assembly) under AS9100/ISO 9001. Builds factory data systems with measured impact: yield loss cut from ~2% to ~0.5% and component fallout from ~5% to under 0.5% on high-volume parts. Supports controlled-impedance builds (stackup and Dk/Df laminate analysis, TDR-verified production), designs KiCad test boards with differential pairs, and coordinates production with customers, suppliers, and international teams (ATPs, conformance reports, shipments).

CORE SKILLS

Manufacturing & Process: SMT assembly, SPI, AOI, selective soldering, conformal coating, equipment commissioning, fixture design, DFM, process optimization

Quality & Standards: AS9100 (internal auditor), ISO 9001, IPC-A-610, root cause analysis (RCA), CAPA, RMA investigation, ATP development, conformance reporting, QMS documentation

Signal Integrity & Test: Controlled-impedance stackups, Dk/Df laminate analysis, TDR impedance coupons, differential pair routing, reflections & impedance matching, transmission line fundamentals, oscilloscopes, spectrum analyzers

Automation & Data: Python (data pipelines, HTML parsing, Gmail API), Power Automate, Microsoft Forms, Excel/VBA, production yield & KPI reporting

Design & Embedded: KiCad, Altium, Fusion 360, Blender, 3D printing, embedded C, ESP32

PROFESSIONAL EXPERIENCE

Process Engineer — PFC Flexible Circuits Ltd. (OSI Systems)

Oct 2024 – Present

Toronto, ON — Flex/rigid PCB manufacturing for aerospace, medical, and security industries

- Support controlled-impedance flex/rigid builds: analyze stackups and laminate datasheets (Dk/Df) alongside senior engineers using field-solver impedance calculators, in a TDR-verified production environment.
- Extract sub-circuits from complex customer designs into standalone KiCad test boards (schematic + layout, including differential pairs) to troubleshoot problem components ahead of production runs.
- Program and commission SPI, AOI, selective soldering, and conformal coating equipment across high-mix SMT and hand-assembly lines; reduced process variability via parameter optimization and custom 3D-printed fixtures (Fusion 360, Blender).
- Built a production yield and revenue-impact reporting system integrating SMT, die-cutting, and functional test data; improvement efforts driven by this reporting cut yield loss from ~2% to ~0.5% on high-volume parts.
- Engineered an automated SMT fallout tracking pipeline (Python, Gmail API, Power Automate) with shift-level trends and weekly executive reports; flagged fallout near 5% on critical parts, enabling actions that cut it below 0.5% on most high-volume parts.
- Deployed a factory-wide maintenance management system (Microsoft Forms + Power Automate + Outlook): digitized 100% of previously untracked verbal requests, creating the facility's first downtime and response-time baselines (~50 requests/month, 12 departments).
- Led architecture and supplier coordination for a facility-wide inventory management ecosystem: universal barcode labeling, PO-validated receiving/kitting scan stations, smart racks, X-ray quantity verification, automated Dynamics 365 posting.
- Coordinate production across operator teams; interface with customers, suppliers, and international/cross-facility teams on ATP derivation and approval, conformance reports, shipment schedules, and material availability.
- Lead root cause analysis on IPC-A-610 non-conformances and customer RMAs with CAPA that eliminated recurring defects; conduct internal audits and author work instructions supporting AS9100/ISO 9001 certification and acquisition due diligence.

R&D and Test Engineering Intern — PFC Flexible Circuits Ltd

Jun 2024 – Oct 2024

- Led design and manufacture of a flex PCB quality-testing device, saving an estimated \$40,000+ in outsourced testing; implemented device communication protocols in C; designed the 3D-printed enclosure in Fusion 360.

SELECTED PROJECTS

Voltage-Controlled Frequency Function Generator

- Designed and built an analog function generator with voltage-controlled output frequency; waveform generation and signal-conditioning stages validated on the bench with oscilloscope measurements.

Non-Intrusive Load Monitoring (NILM) Smart Energy Meter

2023 – 2024

- ESP32-based smart meter detecting individual appliance usage from aggregate consumption via a NILM algorithm; custom PCB, Firebase Realtime Database, and serverless Cloud Functions for automated reporting.

EDUCATION

Bachelor of Engineering, Electrical Engineering — Toronto Metropolitan University

2019 – 2024

- Pierre Lassonde & TMU General Entrance Scholarships. Coursework: Digital Signal Processing, Communication Systems.